

Shraddha Takmoge

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in LinkedIn — 🐙 GitHub — 📁 Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building LLM-powered applications, RAG pipelines, and agentic workflows using Python, LangChain, LangGraph, and MCP. Strong foundation in Machine Learning, NLP, embeddings, Transformers, vector databases, and LLM evaluation, with experience implementing hybrid retrieval and evaluating RAG systems. Built practical AI applications using Pinecone, FAISS, FastAPI, Streamlit, and Docker, with exposure to AWS EC2 and S3.

EDUCATION

Savitribai Phule Pune University

Nov 2022 – Jun 2026

Bachelor of Engineering – Artificial Intelligence and Data Science

Overall CGPA: 9.25 / 10.00

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Machine Learning:** Scikit-learn, Regression, Classification, Feature Engineering, Model Evaluation
- **NLP & Deep Learning:** NLP, Neural Networks, Transformers, Embeddings
- **Generative AI:** LLMs, Prompt Engineering, RAG, Self-RAG, CRAG, LangChain, LangGraph, MCP
- **RAG & Retrieval:** FAISS, Pinecone, Semantic Search, Hybrid Search, BM25, Reciprocal Rank Fusion (RRF)
- **RAG Evaluation:** DeepEval, LLM-as-a-Judge, Precision@K, Recall@K, Hit@K, MRR, NDCG, Context Precision and Recall, Faithfulness, Relevancy
- **Backend & Deployment:** FastAPI, REST APIs, Pydantic, Streamlit, Docker, AWS (EC2, S3), Render
- **Libraries & Tools:** Pandas, NumPy, PyTorch, Hugging Face, LangSmith, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

PROJECTS

RepoLens AI – GitHub Repository Investigator LangGraph — GitHub MCP — RAG — Pinecone — FastAPI

- Built a LangGraph-based repository investigation agent using GitHub MCP tools to inspect repository structure, source files, commit history, and answer repository-specific questions through natural language.
- Implemented hybrid code retrieval using Pinecone dense search and BM25, combining semantic and lexical results with Reciprocal Rank Fusion (RRF).
- Developed an agentic workflow using LangGraph and FastMCP to interact with GitHub through repository analysis, file retrieval, and code search tools.
- Evaluated retrieval on 20 repository-specific queries using Hit@K and Recall@K, achieving 55% Hit@1, 75% Hit@3, 80% Hit@4, and 68.33% Recall@4; removed reranking after evaluation showed weaker retrieval performance.

AI Decision Auditor

Python — LangChain — FAISS — Ollama — Pydantic — FastAPI — Streamlit

- Built a multi-stage decision-auditing pipeline in LangChain, using Pydantic schemas to enforce structured LLM output across decision analysis, evidence verification, and counterargument generation.
- Implemented a retrieval step using FAISS with Ollama embeddings, retrieving the top 3 relevant text chunks per decision and filtering results below a 0.50 relevance-score threshold.
- Designed a rule-based risk-scoring function combining evidence-verification status, missing-information count, and high-importance assumptions into a 0–10 risk score with a low/medium/high rating.
- Exposed the pipeline through a FastAPI backend and Streamlit interface, in addition to a command-line entry point.

EXPERIENCE

Data Science Intern – Rajavi Technology Pvt. Ltd.

Dec 2024 – Jan 2025

- Built a content-based movie recommendation system in Python using the TMDB 5000 dataset (4,800 movies), merging and cleaning movie and credits data with Pandas.
- Converted movie tags into vectors using CountVectorizer (5000 features, stopword removal) and applied stemming with NLTK's PorterStemmer to normalize text.
- Computed cosine similarity between movie vectors to recommend the top 5 most similar movies for a given title, and saved the trained model using Pickle.

ACHIEVEMENTS

- Selected for the **Women in Engineering (WIE) Program** conducted by **Shikshana Foundation** in collaboration with the **AMADEUS** team.
- **Runner-up** at the **ISDM Code4Change Hackathon**